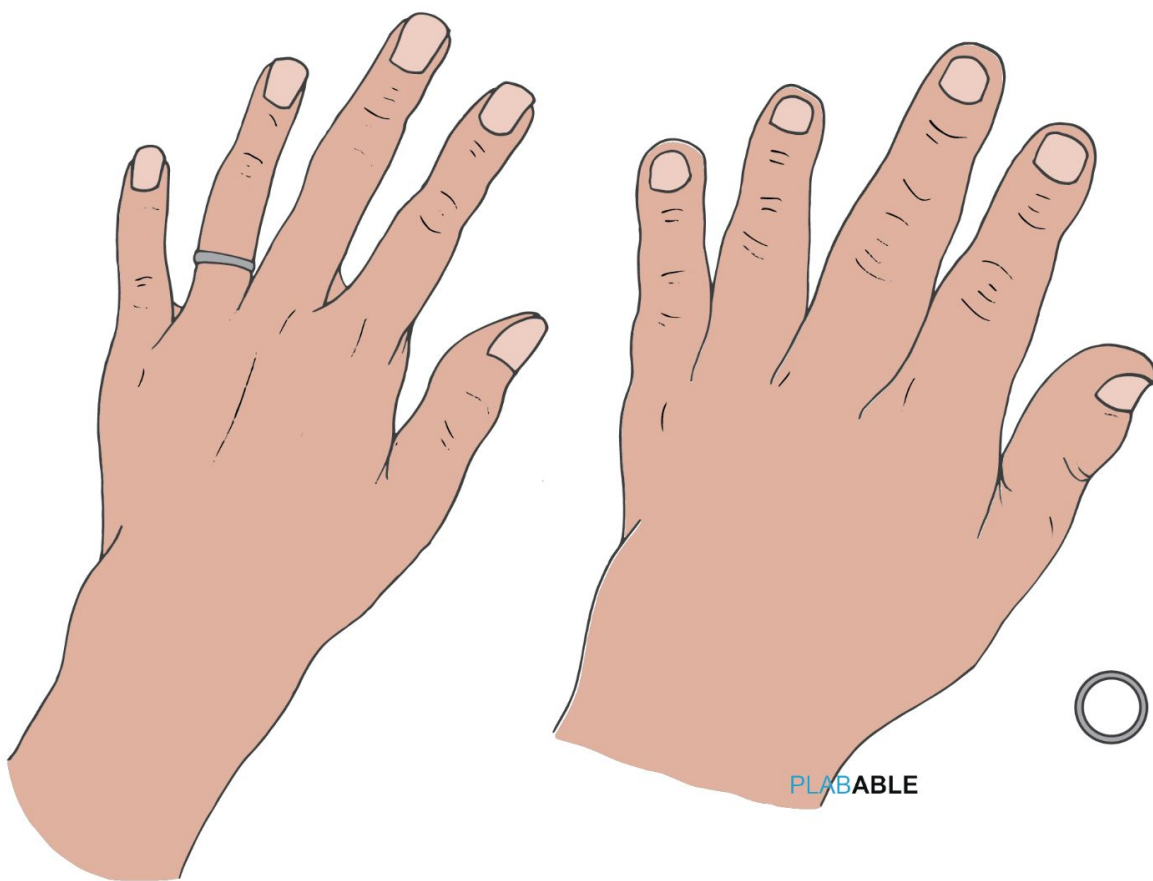


PLABABLE

GEMS 

VERSION 1.1

ENDOCRINE



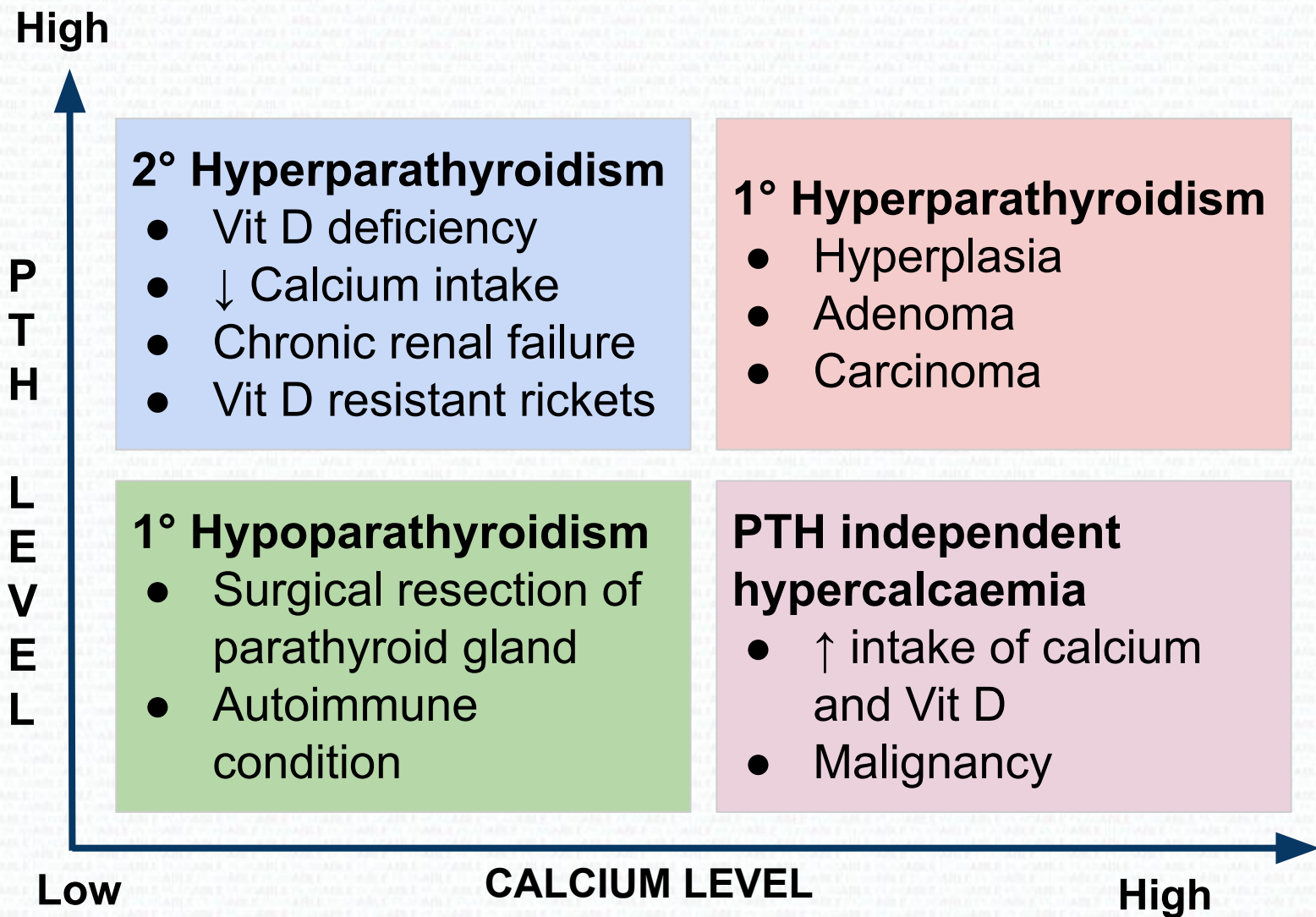
Hypercalcaemia

Mnemonic	Symptoms
Bones	Painful bones especially seen in primary hyperthyroidism
Stones	Renal stones
Groans	● Constipation ● Pancreatitis ● Vomiting ● Nausea
Psychiatric moans	● Lethargy ● Easy fatigue ● Depression

Causes

- **Primary hyperparathyroidism**
- Malignancy - **Multiple myeloma**
- Sarcoidosis
- Hyperthyroidism
- Prolonged immobilization
- Thiazides

Hypercalcaemia



Treatment

- Immediate: **0.9% NS** to increase urinary excretion of calcium
- Loop diuretic: **Furosemide** to increase calcium excretion
- Bisphosphonates: **Zoledronic acid** to reduce bone resorption
- **Denosumab** to reduce bone resorption

Hypocalcaemia

Symptoms “**CATs go numb**”

- Convulsions
- Arrhythmia
- Tetany and carpopedal spasm
- Paraesthesia (perioral, fingers and toes)

Causes

- **Low PTH:**
 - Parathyroid destruction by surgery, metastases or amyloidosis
 - Autoimmune
- **High PTH:**
 - Vit D deficiency
 - PTH resistance - Pseudohypoparathyroidism
- Hyperventilation
- Acute pancreatitis

Hypocalcaemia

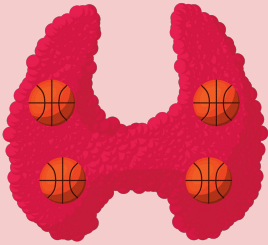
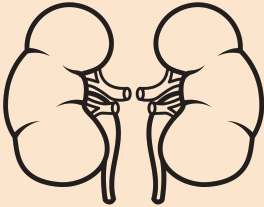
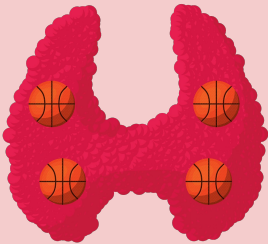
Signs

- **Chvostek's sign** - Tapping of facial nerve
- **Trousseau's sign** - Carpopedal spasm following BP cuff inflation
- Prolonged QT interval

Treatment

- **Symptomatic:** IV calcium gluconate 10%
- **Long term:** Oral calcium supplements with vit D

Hyperparathyroidism

	Cause	Labs	Others
Primary 	Parathyroid adenoma Hyperplasia Carcinoma	↑ Calcium ↓ Phosphate ↑ PTH	Associated with MEN 1 and MEN 2a Treatment is surgery
Secondary 	CKD causes parathyroid hyperplasia due to long standing hypocalcemia	Low-normal calcium ↑ PTH ↑ Phosphate in CKD	Treatment with calcium and Vit D
Tertiary 	Autonomous production of PTH following long standing chronic kidney disease	↑ Calcium ↑ PTH ↑ Phosphate	Surgery is definite treatment Cinacalcet

Diabetes

	Type 1 (insulin deficiency)	Type 2 (insulin resistance)
Cause	Associated with HLA DR3, DR4 and islet cell autoantibodies Prone for ketoacidosis	Risk factors: Obesity, Family history, metabolic syndrome, PCOS, H/O gestational diabetes, and physical inactivity
Diagnosis	Symptomatic patients: Random glucose ≥ 11.1 mmol/L or fasting ≥ 7 mmol/L Asymptomatic patients: Abnormal random plasma glucose or two fasting glucose samples ≥ 7 mmol/L + HbA1c of 48 mmol/L (6.5%) or more	

Diabetes

	Type 1 (insulin deficiency)	Type 2 (insulin resistance)
Treatment	Insulin therapy	Metformin (first line) Dual-therapy and Triple-therapy (if HbA1c >6.5%) of metformin and (DPP-4 inhibitor / sulfonylurea / pioglitazone) If HbA1c is still more than 6.5 - Add Insulin

Diabetes Complications (Acute)

DIABETIC KETOACIDOSIS

- Common in type 1 DM
- Metabolic acidosis and +ve urine ketones
- **Treatment:** IV 0.9% NS, Insulin infusion with dextrose and electrolyte monitoring

HYPEROSMOLAR HYPERGLYCAEMIC STATE

- Seen in type 2 DM
- High plasma glucose level, high serum osmolarity without significant ketoacidosis
- **Treatment:** 0.9% NSI and electrolyte correction

Diabetes Complications (Chronic)

DIABETIC NEPHROPATHY

- Proteinuria
- Microalbuminuria
- Treatment to control HbA1c
- ACEi to prevent progression

INCREASED RISK OF CVD

- Reduce risk by proper glycemic and BP control
- Statins

DIABETIC RETINOPATHY

Microvascular changes:
Aneurysms, hard exudates, haemorrhages, cotton wool spots, and neovascularisation

Treatment:

- **Early:** Glycaemic control
- **Late:** Panretinal photocoagulation, intravitreal steroids and anti- VEGF

Diabetes Complications (Chronic)

DIABETIC NEUROPATHY

- Glove and stocking neuropathy
- Burning, numbness and tingling sensation in the extremities
- Tight glycemic control and prevention of foot trauma

DIABETIC FOOT ULCER

- Glycemic and BP control, Smoking cessation
- Wound management
- Foot interventions to prevent ulcer formation such as well-fitting footwear

Diabetes Ketoacidosis

Presentation

- Polyuria
- Polydipsia
- Dehydration
- Sweet smelling breath
- Altered mental status or even coma
- Most common in **type 1 diabetes**

Signs

- Dry mucous membranes
- Hypotension
- Tachycardia

Precipitating causes: Infection, stoppage of insulin, drugs such as steroids and thiazides

Investigations

- ↑ Blood sugar level (BSL)
- **Positive urine ketones**
- ABG shows **metabolic acidosis** (low bicarb and low pH)

Diabetes Ketoacidosis

Treatment

- **0.9% normal saline**
- **IV insulin infusion** → 5% dextrose when glucose falls below 12 mmol/L
- Hypokalemia → KCl infusion

Complications

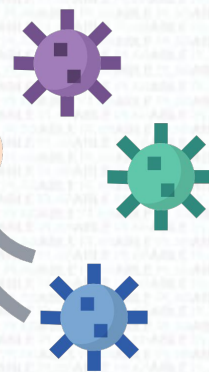
- Cerebral oedema
- Pulmonary oedema
- Cardiac arrhythmia due to electrolyte imbalance



**High
sugar
level**



**Fruity breath
(Acetone)**



MODY

Brain trainer:

A 25 year old non-diabetic presents with mild hyperglycemia. His father and grandmother had some form of diabetes. What is the most likely diagnosis?

→ **Maturity onset diabetes of the young**

Latent autoimmune diabetes of adulthood (LADA)

Latent autoimmune diabetes of the adult (LADA) is a variant of type 1 diabetes which develops much slower than type 1 diabetes.

It develops slowly and often mistaken for type 2 diabetes mellitus since it develops in adulthood.

Test to request for LADA:

GAD antibody testing

This test will help distinguish LADA from type 2 diabetes

Glycosuria

Brain trainer:

A non-diabetic patient after surgery has glycosuria. What is the most likely diagnosis?

→ **Stress hyperglycemia**

Infection or stress → cortisol → hyperglycaemia

Pre-Diabetes

Brain trainer:

What laboratory values are found with impaired glucose intolerance?

→ Fasting → 5.5 to 6.9 mmol/l

→ 2 hr post-prandial → 7.8 to 11.0 mmol/l

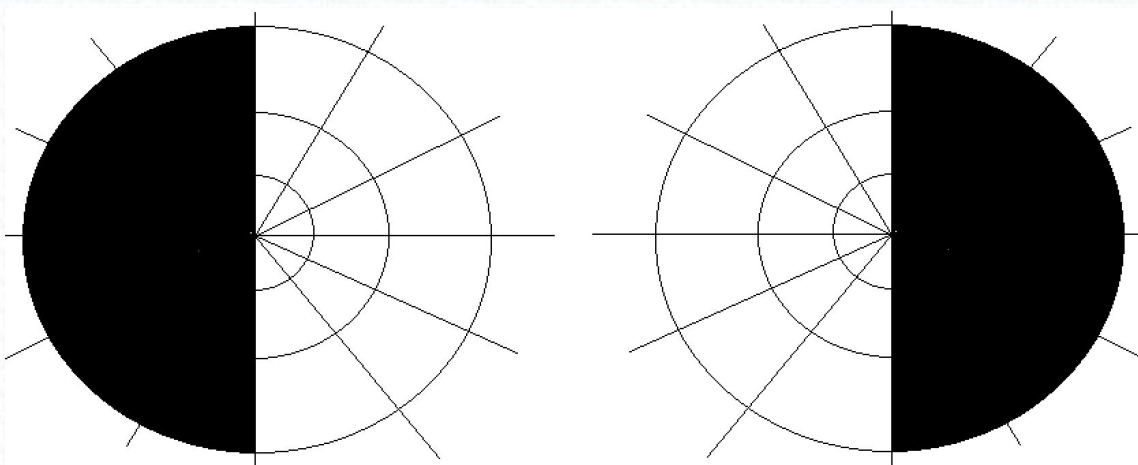
Hyperprolactinaemia

Causes

- **Prolactinoma**
- Hypothyroidism
- Drugs: antipsychotics (risperidone & haloperidol) and domperidone
- Brain injury

Presentation

- **Females:** Inhibits FSH and LH causing menstrual irregularity (amenorrhea) and galactorrhoea
- **Males:**
 - Secondary hypogonadism
 - Reduced libido
 - Gynecomastia
 - Erectile dysfunction
- **Visual symptoms:** Bilateral hemianopia
- Headache

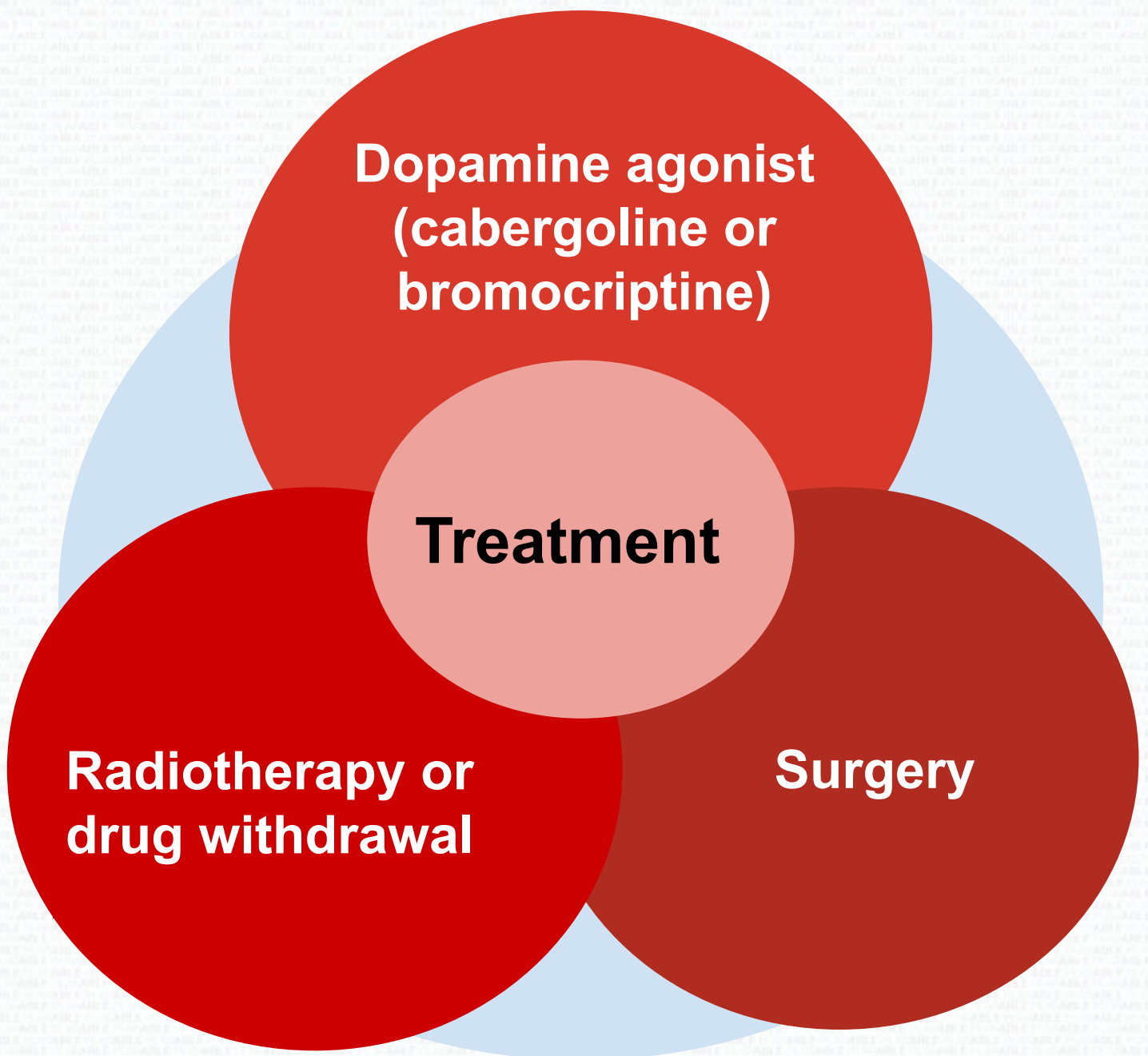


Bitemporal Hemianopia

Hyperprolactinaemia

Investigations

- ↑ serum prolactin level
- Pituitary MRI
- Visual field testing



Thyroid Disorders

Hypothyroidism	Hyperthyroidism
Decreased appetite but weight gain	Increased appetite and weight loss
Cold intolerance	Heat intolerance
Constipation	Diarrhoea
Bradycardia	Palpitations and tachycardia
Menstrual irregularities - Menorrhagia	Menstrual irregularities - Oligomenorrhea
Tiredness and lethargy	Irritability and weakness
● Deep hoarse voice ● Reduced libido	● Tremor ● Increased sweating ● Brisk reflex ● Lid lag ● Palmar erythema

Thyroid Disorders

	Hypothyroidism	Hyperthyroidism
Causes	Primary: (low T4 and high TSH) Autoimmune - Hashimoto's thyroiditis Iodine deficiency Drugs - Amiodarone and antithyroid medications Congenital hypothyroidism Secondary: (Low TSH and T4) ● Hypopituitarism ● Isolated TSH deficiency	Graves disease Antimicrosomal antibodies against thyroid peroxidase and Antithyroglobulin antibodies Toxic nodular goitre De Quervain's thyroiditis Exogenous thyroid ingestion

Thyroid Disorders

	Hypothyroidism	Hyperthyroidism
Treatment	Thyroxine (T4) supplementation If due to iodine deficiency then replacement of iodine	Anti-thyroid drugs Carbimazole Bone marrow suppression and cutis aplasia (pregnancy) Propylthiouracil Liver failure Radio-iodine therapy Thyroidectomy

Hyperthyroidism (pregnancy)

Pre-pregnancy and first trimester:

- Propylthiouracil

2nd, 3rd trimester + post-pregnancy

- Carbimazole
- Partial thyroidectomy (in 2nd trimester if carbimazole is not effective)

Notes:

- Carbimazole has risk of aplasia cutis and omphalocele in the fetus.
- Propylthiouracil: risk for hepatotoxicity in the mother
- Lowest possible dose should be prescribed
- Radioiodine therapy contraindicated

Thyroid Crisis

Brain trainer:

What is the management for thyroid storm?

- Propylthiouracil
- Palpitations → beta-blocker
- If infective etiology → broad-spectrum IV antibiotics

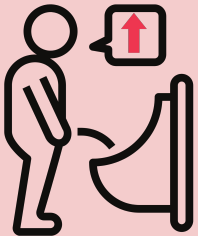
Postpartum Thyroiditis

Brain trainer:

What medication is indicated to control the symptoms of postpartum thyroiditis?

→ **Beta-blocker (palpitations + sweating)**

ADH Disorders

	Diabetes insipidus (decreased ADH)	SIADH (excess ADH)
Symptoms	Polyuria (>3L of urine/day) Polydipsia and thirst 	
Causes	Cranial: Decreased secretion of ADH ● Craniopharyngioma ● Head injury ● Sarcoidosis and tuberculosis ● Infections: Meningitis and encephalitis ● Post-radiotherapy Nephrogenic: ADH resistance ● Lithium ● Renal tubular acidosis ● CKD ● Idiopathic	Small cell lung cancer Meningitis and encephalitis Drugs: Chlorpropamide Oxytocin Amitriptyline

ADH Disorders

Urine Osmolality	Cranial DI	Nephro-genic DI	Psychogenic polydipsia
Normal	<300	<300	<300
Water deprivation	<300	<300	>800
Desmopressin injection	>800	<300	>800

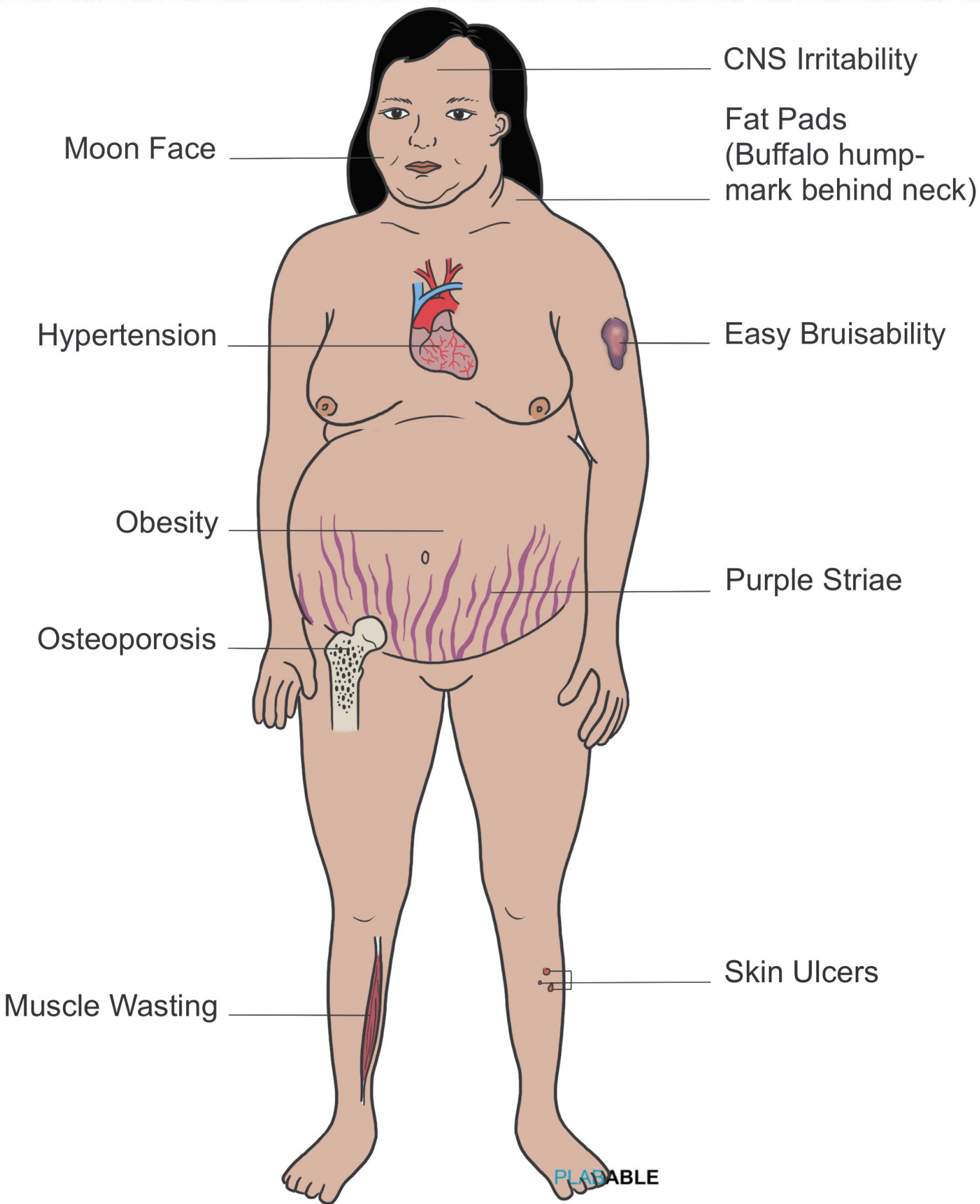
Treatment	
Diabetes Insipidus	SIADH
Central: Desmopressin Nephrogenic: High dose desmopressin with or without thiazide and NSAIDs	● Fluid restriction (mild cases) ● Demeclocycline ● Conivaptan and tolvaptan

Cushing’s Syndrome

Prolonged exposure to either endogenous or exogenous cortisol

ACTH dependent	ACTH independent
Cushing’s disease (ACTH from pituitary)	Adrenal adenoma
Ectopic ACTH producing tumours	Adrenal carcinoma
Excess ACTH administration	Exogenous glucocorticoid administration

Cushing's Syndrome



Cushing's Syndrome

Investigation

To confirm cushing's syndrome:

- 24-hour urinary free cortisol
- Low-dose dexamethasone suppression test

To identify the cause:

- Plasma ACTH: High - ACTH dependent
- Plasma ACTH: Low - ACTH independent

If ACTH is high to differentiate between pituitary and ectopic cause:

- High-dose dexamethasone test
- MRI pituitary

If ACTH is low:

CT or MRI scan of abdomen to look for adrenal tumour

Treatment

- **Medications** to inhibit cortisol synthesis
 - Metyrapone
 - Ketoconazole
 - Mitotane
- **Surgery** - Pituitary tumour
- Removal of exogenous corticosteroid

Pheochromocytoma

High levels of catecholamine production

Symptoms and signs

- Palpitations
- Profuse sweating
- Anxiety
- Sense of doom
- Headache
- Tremor
- Hypertension

Investigation

- 24-hour urinary metanephrines and VMA
- CT abdomen to locate the tumour

Associated with

- Multiple endocrine neoplasia (MEN 2A and 2B)
- Neurofibromatosis
- Von Hippel-Lindau syndrome

Treatment

Surgery after preoperative alpha-blocked with phenoxybenzamine

Adrenal Insufficiency

Primary adrenal insufficiency/Addison's disease

- Autoimmune cause
- Infections and haemorrhage
- Congenital adrenal hyperplasia
- Drugs: ketoconazole

Secondary adrenal insufficiency

- Hypothalamic and pituitary failure
- Long term steroid medication

Clinical features

Acute: Hypotension, shock, abdominal pain and vomiting

Chronic: Fatigue, weakness, weight loss, salt cravings, syncope, vomiting, and hyperpigmentation in buccal mucosa and lips (primary)

Labs

- **Low sodium and high potassium**
- Early morning cortisol - Low
- ACTH: high in primary and low in secondary
- ACTH stimulation test: Cortisol does not rise

Treatment

- Hydrocortisone (Glucocorticoid)
- Fludrocortisone (Mineralocorticoid)

Addisonian Crisis

Causes

- Withdrawal of chronic steroid therapy
- Infection or stress

Clinical features

- Shock (confusion, postural hypotension, tachycardia and oliguria)
- Abdominal pain (may be severe enough to mimic an acute abdomen)
- Hypoglycaemia

Labs

- Cortisol and ACTH level
- FBC, U&Es, blood glucose
- Cultures (blood/urine/sputum)

Treatment

- IV fluids if shocked
- IV hydrocortisone 100mg
- Hypoglycaemia → IV glucose infusion
- If the condition improves in 72 hours → Switch to oral steroids
- If an adrenal pathology is identified → Fludrocortisone may be indicated

Hyperaldosteronism

Primary hyperaldosteronism or Conn's syndrome:

- Adrenal adenoma
- Adrenal hyperplasia
- Adrenal carcinoma

Secondary hyperaldosteronism

- Renin producing tumour
- Renal artery stenosis

Clinical features

- Hypertension
- Hypokalemia
- Metabolic alkalosis

Investigations

- Renin and aldosterone levels
- CT/MRI
- Adrenal venous sampling (gold standard for primary)

Treatment

- Aldosterone antagonist - spironolactone
- Surgery: adrenalectomy

Sheehan Syndrome

Postpartum hypopituitarism due to pituitary necrosis as a result of intra-partum or post-partum haemorrhage causing hypotension

Symptoms

- Absence of lactation
- Amenorrhea or oligomenorrhea
- Hypothyroidism
- Adrenal insufficiency

Labs

- Low TSH and T4
- Low ACTH and cortisol
- Low LH and FSH
- Low estrogen

Treatment is with specific hormone replacement

Note: Pituitary apoplexy due to pituitary tumour presents with acute headache, visual field defect and symptoms of hormone deficiency

Amenorrhoea

Primary: Never had menses till 13 years without secondary sexual characteristics and 15 years with secondary sexual characteristics

Causes

- Constitutional delay
- Ovarian failure
- Hypothalamic failure
- Kallmann's syndrome and Imperforate hymen
- Congenital adrenal hyperplasia

Secondary: Absence of menses for 6 months in someone who have had menses before

Causes

- Pregnancy and lactation
- Menopause
- Premature ovarian failure and PCOS
- Pituitary and hypothalamic disease, and hyperprolactinemia

Amenorrhoea

Investigation

- **Primary ovarian failure:** FSH and LH increased
- **PCOS:** LH:FSH ratio is increased
- **Hyperprolactinemia:**
 - Pituitary tumour (MRI)
 - ↑ Serum prolactin level
- **Hypothyroidism:**
 - Low T3
 - Low T4

Treatment

- Hormone replacement therapy - ovarian failure
- Treatment focussed on specific condition

Acromegaly

↑ **Growth hormone** secretion due pituitary tumour

Clinical features

- Enlargement of hands and feet
- Frontal bossing
- Macroglossia
- Prognathism (enlargement of jaw)
- Coarse facial features
- Hypertension, cardiomyopathy and arrhythmias
- Type 2 diabetes mellitus



Acromegaly

Labs

- **IGF-1 (screening)**
- Oral glucose tolerance test to confirm
- MRI scan of pituitary

Treatment

- **Trans-sphenoidal surgery** is the treatment of choice
- Somatostatin analogues - octreotide
- GH receptor antagonist - pegvisomant
- Radiotherapy

Metabolic Acidosis

Arterial pH <7.35 and plasma bicarbonate <22 mmol/L

H⁺ excess:

- Ketoacidosis - Starvation and DM
- Lactic acidosis - Heart failure, Drugs
- Methanol, Salicylate and Ethylene glycol poisoning
- Renal failure and Type 1 RTA

Loss of bicarbonate:

- Diarrhoea
- Type 2 RTA
- Drugs: acetazolamide

Respiratory compensation immediately by hyperventilation

Renal compensation by increased bicarbonate absorption or H⁺ excretion

Metabolic Alkalosis

Arterial pH > 7.45

Causes

- Vomiting (loss of H⁺)
- Hyperaldosteronism
- Hypokalemia (H⁺ shifts outside and K⁺ shifts inside the cell)
- **Respiratory compensation** occurs immediately and involves decreased respiratory rate to increase PCO₂
- **Renal compensation** occurs late and involves increased excretion of bicarbonate

Respiratory Acidosis & Alkalosis

Respiratory acidosis	Respiratory alkalosis
Hypoventilation (PaCO2 > 45 mmHg)	Hyperventilation (PaCO2 < 35 mmHg)
- Severe asthma or COPD - Suppression of respiratory center by drugs such as opioids - Obesity hypoventilation syndrome	- Panic attack - Aspirin toxicity (mixed respiratory alkalosis and metabolic acidosis)

Pseudogout

Brain trainer:

A 29 year old woman presents with a swollen, painful right knee joint. She also complains of constipation, feeling cold and weight gain. Joint aspiration of her knee joint will show what finding?

➔ **Positive birefringent crystals**

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